Huawei Access Node KOR-ACC-83 Optical Transport Failure SOP (ALARM_2973)

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2973

Description

The Optical Transport Failure failure domain on Huawei Access Node KOR-ACC-83 at Koramangala Datacenter is a complex phenomenon that arises from the intricate interplay of physical, electrical, and optical parameters on the SFP/XFP interface. As a crucial component of the optical transport layer, the SFP/XFP module is responsible for transmitting and receiving optical signals between the node and the fiber-optic cable. The degradation of the SFP/XFP module or the fiber cut can lead to a significant reduction in the received power, resulting in optical transport failure. Furthermore, the Huawei Access Node's robust architecture and sophisticated control plane mechanisms, including the Optical Transport Network (OTN) protocol, can exacerbate the issue, making it challenging to diagnose and resolve.

Pre-requisite Checks

- Check: Verify that the Koramangala Datacenter is accessible and the site is not experiencing any power or environmental issues., Action: Dispatch remote hands to the site to perform the checks.

- Check: Verify that the redundant link is functioning properly and is not experiencing any issues., Action: Check the link status and perform any necessary actions to ensure redundancy is functioning correctly.

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display otn alarms, Expected Output: OTN Alarm Information, Description: This command will display the current OTN alarms, helping to isolate the issue.

Command: display interface GigabitEthernet 0/1/0, Expected Output: Interface Information, Description: This command will display the interface information for the GigabitEthernet 0/1/0 interface, including the SFP/XFP status and link status.

Command: display optical power GigabitEthernet 0/1/0, Expected Output: Optical Power Information, Description: This command will display the optical power information for the GigabitEthernet 0/1/0 interface, helping to identify any issues with the SFP/XFP module or fiber.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Module, Command: replace wdm-sfp 0/1/0, Description: This step will replace the SFP/XFP module on the GigabitEthernet 0/1/0 interface, resolving the optical transport failure.

Step 2: Step: Repair or Replace the Fiber, Command: repair fiber GigabitEthernet 0/1/0, Description: This step will repair or replace the fiber-optic cable on the GigabitEthernet 0/1/0 interface, resolving the optical transport failure.

Post-Check Verification

```
Command: display otn alarms, Expected Output: No OTN alarms, Description: This command will verify that the OTN alarms have been resolved.

Command: display interface GigabitEthernet 0/1/0, Expected Output: Interface Up, Description: This command will verify that the interface is up and functioning properly.

Command: display optical power GigabitEthernet 0/1/0, Expected Output: Optical Power Normal, Description: This command will verify that the optical power is within normal limits.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: undo replace wdm-sfp 0/1/0, Description: This step will restore the original SFP/XFP module on the GigabitEthernet 0/1/0 interface, reverting to the previous configuration.

- Step: Restore the Original Fiber, Command: undo repair fiber GigabitEthernet 0/1/0, Description: This step will restore the original fiber-optic cable on the GigabitEthernet 0/1/0 interface, reverting to the previous configuration.